Harsha Vardhan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.67
Fiitjee Junior college Saifabad	Class XII (Telangana State Board of Intermediate Education)	2023	85.20%
ST. Peters High School Dhullapally	Class X (CBSE)	2021	80.00%

POSITIONS OF RESPONSIBILITY

Inter IIT Hospitality Coordinator:

Coordinated hospitality and accommodation-related operations for approximately **2,300-2,400** participants and **2,000-2,100** staff members during Inter IIT. Managed communication, logistics, and the smooth execution of hospitality activities.

Diesta Hospitality Head:

Led the hospitality team for Diesta, overseeing guest management, accommodation, coordination, and overall hospitality operations throughout the event.

Milan Production & Pronites Head:

Led the production and Pronites operations for Milan, coordinating stage production, technical requirements, artists, and the execution of major event nights for a fest with a footfall of approximately **2,000-3,000** attendees.

Otaku Club Overall Coordinator:

Serves as the Overall Coordinator of Otaku Club, overseeing the club's activities, team coordination, event planning, and initiatives related to anime, manga, and Japanese culture.

SN Bose Hostel HR:

Worked as the HR representative for SN Bose Hostel, contributing to team coordination, member management, and the smooth functioning of hostel-related activities.

SKILLS

Technical Skills: Python, SQL, API Integration, Vibe Coding, Prompt Engineering, 3D Modeling (Sloyd Edge), Excel, Google Sheets, Power BI, Git, GitHub.

Leadership & Professional Skills: Leadership, Team Management, Communication, Cross-functional Coordination, Stakeholder Management, Operations Management, Problem Solving, Crisis Management, On-ground Decision Making.

RELEVANT COURSES

Mathematics : Calculus-I, Calculus-II, Elementary Linear Algebra, Differential Equations.

Programing : Introduction to Programming, Chemical Process Calculations, Applied Mathematics in Chemical Engineering, Numerical Methods, Computer Vision in Chemical and Life Sciences, AI/ML Tool Chains for Chemical Engineers.

PROJECTS

InsightAI -- AI-Powered Decision Intelligence Platform(github.com/siddu727/InsightAI**) :**

- Built a multi-agent pipeline that converts raw Excel/CSV datasets into verified, executive-ready reports -- an LLM planning agent (Google Gemini) selects relevant analyses based on the dataset's schema, while a deterministic Python/pandas engine executes the actual computations, preventing hallucinated numbers.
- Designed a **scalable, registry-based architecture** where each analysis type (trend, growth, Pareto, anomaly detection) is a self-contained, pluggable "skill" -- new analytical capabilities can be added independently without modifying the core pipeline, ingestion layer, or agents.
- Implemented a schema-inference layer that adapts the AI's investigation plan dynamically to whatever dataset it's given, instead of relying on hardcoded, dataset-specific logic -- making the system generalizable across business domains.
- Built a second LLM agent to synthesize computed insights into a structured executive narrative, cleanly separating data analysis from business communication.
- Used Pydantic for typed data models across the pipeline (schema, plan, insights), ensuring consistent, validated data flow between the ingestion, planning, execution, and reporting stages.
- Developed an interactive Streamlit interface with file upload, live progress tracking, and dual export (markdown report + JSON insights), alongside a scriptable CLI entry point.
- Tech stack: Python, pandas, Streamlit, Google Gemini API, Groq API, Open router API, Pydantic, openpyxl.

Computational Modeling & Numerical Methods -- Interactive Solver & Visualizer(github.com/siddu727/Computational-Modeling-Numerical-Methods**) :**

- Built a full-stack interactive application (FastAPI + React/Vite) that implements and visualizes advanced numerical methods -- Gauss-Legendre quadrature via the Golub-Welsch eigenvalue algorithm, LU decomposition, Newton-Raphson root-finding, companion-matrix methods, and spectral/collocation ODE solving.
- Designed a modular backend where each numerical method is a self-contained Python module dynamically loaded and exposed as an independently testable, typed REST endpoint (FastAPI + Pydantic validation).
- Implemented high-precision numerical routines using Decimal/Fraction arithmetic to avoid floating-point error in high-order polynomial coefficient computation -- a detail most implementations skip.
- Built a React frontend with a dedicated interactive visualization tool per method, letting users adjust parameters (polynomial order, quadrature points, ODE domain) and see results update live.
- Set up CI/CD with GitHub Actions to automatically lint/build the frontend and validate the backend on every push and pull request.
- Tech stack: Python, FastAPI, NumPy, SciPy, Matplotlib, React, Vite, Tailwind CSS.

EXTRACURRICULAR

Sports & Interests : Hockey, Badminton, Basketball, Recreational Multi-Sport Participation, Anime & Japanese Culture.